

Architecture Guidelines For In-company Platform and Technology Dept: Preliminary Study

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ABSTRACT

The present-day scenario of accelerated digital transformations has led industries such as financial institutions to increase their "in-company technology depth" significantly with hybridizations of AI, HPC computing, and distributed cloud infrastructures. However, with increased technological complexity comes a critical gap between localized performability and ESG compliance. The "dark side of cloud provisioning" with high energy costs and lack of visibility of resource utilization has emerged as a major operational risk with organized enterprises and appears to impact ESG compliance. This preliminary study is the starting point for the development of a Statement of Architectural Work (SoAW) for in-house computing guidelines. By employing a systematic mapping approach for the analysis of relevant research and industry best practices (between 2016 and 2025) for green architectural strategies and service-level energy modeling and infrastructure lifecycle analysis, we set out to determine the design patterns and measurement frameworks with proven performance and the capability for implementation within a high-compliance setup.

The analysis exposes that the traditional infrastructure-oriented parameters, such as CPU usage, are not beneficial in understanding complex multi-tenant and microservices-oriented environments. Substantial 'hidden' costs in terms of energy use were found in inter-container communication networking and secondary monitoring services, which are not yet considered in terms of accounting. Moreover, it has been observed, according to the findings, that there is no uniformity in implementing energy-efficient design patterns in different technology departments. The major result is a reusable classification method which groups architectural guidelines into four strategic fields: Software Design Efficiency, Digital Sufficiency in Infrastructure, Industrial Operationalization, and Holistic Governance. It should be noted that the classification method provides the technical basis needed to write the formal Statement of Architectural Work. The discussed guidelines are intended to consolidate the concept of "carbon-aware computing" into a common architectural need in order to offer Enterprise Architects a roadmap in which they can ensure a matching level of depth and rigorous governance.

KEYWORDS

Architectural Guidelines, In-house Technology Depth, Sustainable Enterprise Architecture, Statement of Architectural Work (SoAW), Cloud-Native Microservices, Green Computing Tactics, ESG Integration in IT, Operational Risk Management

1 INTRODUCTION

The quick pace of digital transformation has compelled contemporary business to increase their "technology depth," referring to the complexity and sophistication of their corresponding software foundations and cloud platforms. Nevertheless, in today's increasing assimilation of emerging technologies in their ecosystems, including Artificial Intelligence and High-Performance Computing, a paradox arises for modern business, wherein technical excellence is conflicting with an essential contemporary principle of sustainability. Industry trends clearly underline that, in most contexts, performance continues to be prioritized, even though its related environmental and financial "hidden costs," in regards to their complex architecture, continue to be untouched to date [16].

The internal technology divisions within companies are currently faced with managing a multi-tenanted, distributed ecosystem that calls for both high availability and energy conservation. Research suggests that architectural-level decisions, varying from recommendations on using REST technologies to service communication within a cloud-native setup, heavily influence an organization's carbon footprint [5]. Despite the existing strategies within the field of green architectures [17] being made accessible, existing implementation strategies face a substantial barrier. This document proposes a direct survey of existing implementation strategies. In addition, the paradigm shift towards "Green AI" and green computing is only scarcely optimized by optimized code and, hence, strongly asks for a global management strategy. The integration of cognitive computing and cyber-physical systems into Enterprise Architecture Management (EAM) implies, hence, that sustainability should be integrated into future Architectures' best practice. Without well-structured evidence-based best practice, however, it may be jeopardized by lack of inter-operability and global management architectures.

2 BACKGROUND

The intersection of financial technology with environmental sustainability has created a need for commonly accepted architectural guidelines. Therefore, this section outlines the basic concepts that underpin Green IT and Platform Engineering, subsequently underpinning the understanding of the proposed methodology.

A. Sustainable Software Engineering : Sustainable Software Engineering is an emerging field that studies the design and construction of applications regarding their carbon effectiveness. According to the Green Software Foundation, there are three key dimensions with which we can describe sustainability in software: carbon efficiency, energy efficiency, and carbon awareness[8]. In financial

applications that typically exhibit high requirements for availability and low latencies, developers tend to favor an attitude toward performance rather than optimizing for resources. Knowledge of the carbon footprint within coding logic algorithmic complexity and the frequency of transferring data—develops an architectural baseline.

B. Green Platform Engineering and Infrastructure: Platform engineering is an internal service provider for tools and workflows aimed at enabling developer self-service. Under the perspective of sustainability, even the choice of hosting environment—on-premises data centers vs. various flavors of the PCPs carries a huge risk in terms of environmental overhead. The relevant metrics in this regard are:

- Power Usage Effectiveness (PUE): A ratio describing how efficiently a data center uses energy.
- Carbon Intensity: The measure of how much carbon is emitted per kilowatt-hour of electricity consumed, which varies by geographical region and time of day[8].

Architectural guidelines for platform teams involve selecting "Green Regions" and implementing auto-scaling policies to minimize "zombie" resources that consume power without providing utility.

C. The Financial Sector and ESG Compliance: Financial institutions are getting increasingly wrapped in ESG compliance. Most attitudes towards IT in banking traditionally have had a focus on "Governance." "Environmental," on the other hand, is the one that requires movement at a technical operational level. The current standards within industries, such as the GHG Protocol, categorize emissions within three scopes. It is within the Technology division that the greatest impact is usually found on Scope 2 (emissions from electricity consumed but not owned) and on Scope 3, covering most emissions related to value chains, such as those for cloud service providers[14].

D. The Gap in Current Architectural Practices: Initial observations within the industry suggest that there is some disconnection between the high-level ESG ambitions and the day-to-day workings of technology teams. For instance, there will often be no tooling available to developers that would enable them to assess the possible carbon impact of their deployments. In other words, sustainability, as an NFR, would not be present in any decision frameworks provided to platform engineers, which instead focus on security, scalability, and cost.

3 RELATED WORK

The situation with the tech departments in enterprises has reached a point where it is vital to marry local performance requirements with globally considered sustainability challenges. This demands that software architecture be addressed from a multi-dimensional standpoint that goes beyond being energy efficient and encompasses social, economic, and technical aspects related to sustainability. [6] argued that cloud-based solutions today are too opaque, requiring a framework informed by sustainability perceptions that help address multiple aspects at the design level. A key area in this architectural

development journey is the rise of what is being coined as "Green AI" and environmentally sustainable software architecture. Some research make a contribution to the community by synthesizing a body of green architectural strategies, terming them as high-level design strategies specifically designed with the aim of reducing the energy impact of ML-infused systems. However, in a similar area and complementing the one previously highlighted is [10], who proposed the ENRICO system. It does this by directly incorporating energy analysis within the CI/CD pipeline and allowing the developer to have real-time analysis and suggestions to decrease the energy impact associated with newly developed system functionality. The choice of execution environments and frameworks is also an important consideration in regards to organizational technology depth. The work of [5] goes on to show that resource consumption is highly dependent on REST frameworks/execution environments, offering an organizational technology guide that balances performance with optimized resources for firms to choose their technology according to this prescription, thus satisfying an organizational need for standardized operation procedures for in-house departments of firms.[4] introduces an open architecture for an enterprise architecture that promotes incorporating new frontier technologies such as AI/High Performance Computing to avoid vendor lock-in effects while achieving sustainable organizational flexibility for high availability with new HW approaches. Infrastructural management, specifically with regard to data centers, continues to be the bedrock of sustainable compute infrastructure. [13] recommend the use of Life Cycle Assessment, specifically with regard to edge data centers, pointing out the radical impact that the use of renewable energy and recycled servers can make on compute energy use. [13] continue the infrastructural theme with regard to management, though [9] offer a new term, "digital sufficiency." Their paper makes it clear that not only should architectural standards make use of efficiencies but must also make use of user preference management to decrease the energy footprint of data centers. At last, for such a complex technological ecosystem, there also have to be strong guidelines for regulation and management. In a bid to regulate massive tech development, there are some guidelines based on Actor-Network Theory and Enterprise Architecture Management. These guidelines state clearly that for the effectiveness of new communications technologies such as Big Data and IoT, there has to be a holistic regulation of the ecosystem.[11] give a design insight into the distributed computing industry ecosystem and again emphasize the importance of strong and verified processes for ensuring the long-term tech sustainability of the processes. Although the paradigm shift towards "green" architecture has been well recorded in history, in the application domain of industry-scale platforms for industries, there are certain hidden costs that accompany the implementation of this new paradigm.

According to [16], the "dark side of cloud & edge computing" includes a lack of sustainability concerns in current research. The exploratory research reveals that in current trends, organizations seek short-term financial gains rather than long-term sustainability. There exists a burning need for "countermeasures" in pattern-based models for sustainable trends in the provisioning of the cloud. In this direction, [17] argue that current consumers in the cloud lack

transparency in understanding the influence of industry-scale platforms' architectural designs related to financial gains in terms of energy consumption. In this regard, they suggest a reusable list of industry-scale strategies specifically designed for public clouds that would enable organizations to create "energy-efficient by design workloads in the public cloud." There are also benefits in translating theory into practice with the support of existing evaluations made on these techniques in an industry setup. For instance, [7] carried out experimental evaluations on energy efficiency techniques in various companies to prove that despite the efficiency offered by techniques like row/page compression in the database, their natural fit in industry practice is still an area worthy of great consideration in their programs. This study shows that evidence-based advice on choosing techniques is much needed rather than depending on generic techniques in industry practice. [1] claim that information system architectures need to be studied through the prism of technological and social perspectives in relation to information technology staff by ensuring their guidelines in organizational structures remain flexible enough to adapt to different demands at different levels. One of the most important hurdles in applying sustainability principles lies in the inability to measure it using current tools. According to [18], traditional system metrics such as CPU usage cannot remain applicable in cloud-native systems that support multi-tenancy and complex distributed systems. To address this challenge, the authors have proposed a comprehensive automated experimentation methodology in which developers can easily check energy-efficient behavior at all levels of a Kubernetes-based system, starting from hardware to the service system of the application. There is a huge gap in service-level energy modeling in sustainability studies. They specifically stated that most of the current studies lack consistency in estimating energy consumption of distributed interaction mechanisms such as cross-container interaction, network traffic, or even auxiliary services such as monitoring and observability systems.

Finally, the incorporation of these tenets into more general management of an enterprise is considered with a focus on collaborative networks and digital transformation. According to [3], there is a recognized need for a focus on interoperability and generalized frameworks if a sustainable adoption of this type of new technology is going to be possible in a digital economy. A closely allied emphasis is expressed in a later work by [15], in which this author clearly states that "bridging the gap" between AI and reality is only possible if sustainability becomes a key item in new architectural guidelines going forward. Failing this, a degree of progress in areas such as high-performance computing, considered in a work such as [2] piece on exascale computation, will have had an unsustainable positive or "green" aspect due to its adverse impact upon the environment. A sure key item in these architectural guidelines will be incorporating cognitive computing and cyber-physical systems [12] to help develop a "carbon-aware" in-company technology platforms.

4 STUDY DESIGN

4.1 Research Goal

The proposed initial study aims to perform an analysis of the current workflows for their utilization and provisioning of compute resources in order to establish architectural guidelines with an emphasis on sustainability. The investigation centers on the operationalization of carbon-aware energy-efficient practices across the complete compute lifecycle and developer and platform engineering decisions within the technology department of a financial institution.

4.2 Definition Goal

Following the GQM approach, the research goal is formally defined as follows:

Object of Study: Compute infrastructure lifecycle, encompassing resource provisioning, utilization patterns, and platform selection workflows.

Purpose: To understand existing inefficiencies and define a preliminary set of sustainable architectural guidelines focusing on compute.

Quality Focus: Integration of environmental sustainability as a primary Non-Functional Requirement (NFR).

Stakeholders: Software/Application developers, platform engineers, and infrastructure architects.

Environment: A high-availability corporate financial technology department.

4.3 Research Questions

In trying to bridge the gap that has emerged between current technology operations and institutional objectives on sustainability, this preliminary study has been guided by two major research questions. These relate to current technical definitions, operational at the financial institution, and that which needs to be incorporated relative to the concepts of sustainability into the compute lifecycle.

- **RQ1:** How can we define and implement compute architecture guidelines that improve sustainability? This question attempts to understand the gap between traditional compute provisioning and a carbon-aware architecture. It focuses on identifying specific technical metrics such as Carbon Intensity and Power Usage Effectiveness (PUE) required to codify sustainability into the organization's standard operating procedures. The investigation seeks to understand how "carbon-aware" scheduling and resource allocation can be applied without compromising the strict high-availability and low-latency mandates that characterize financial services.
- **RQ2:** What architecture principles must be operationalized in compute guidelines? This question considers how to move from "performance-centric" thinking to an approach in which sustainability is integrated into the operational mindset of the organization. It explores the operationalization of specific principles, including:

- **"Good Enough" Architecture and Right-sizing:** Implementing rigorous right-sizing strategies for CPU and memory allocation. This involves shifting away from "peak-load provisioning" to a model where compute resources are matched precisely to actual workload requirements, thereby eliminating "zombie" energy consumption and idle resource waste.
- **Reuse-before-build:** Prioritizing the optimization of existing multi-tenant clusters and shared services over the provisioning of new, dedicated virtual machine instances that contribute to infrastructure sprawl.
- **Secure and Sustainable by Design:** Investigating how intensive financial security and compliance protocols (eg, encryption-at-rest, real-time auditing) can be executed with maximum energy efficiency, ensuring that security integrity does not result in unnecessary computational overhead.

In this regard, therefore, the study places the compute environment at the focal point of detail diagnosis while opening pathways toward more sustainable operational models. The journey between RQ1, which postulated the existence of technical metrics, and RQ2, in which architectural principles are practically applied, has allowed sustainability to be accounted for as a de facto non-functional requirement and not some elective feature. It is in this way that these investigations will provide the evidentiary base of the architectural guidelines at their preliminary interstitial, linking high-level ESG mandates with daily technical workflows and platform engineers' and developers' activities. The structured manner in which this will be done will lend credence to the fact that the resulting guidelines will indeed be technically viable and consonant with the required financial services sector standards of performance.

4.4 Initial search

The initial search for data was twofold: One is a systematic literature search across the primary academic databases, while the other is an internal empirical study within a large-scale financial institution. This will ensure that the resulting compute guidelines are underpinned by both academic and industry-specific constraints.

A. Academic Database Search: A systematic search across the three principal digital libraries was conducted to identify any extant frameworks and architectural patterns. The search period was between 2016 and 2025 to capture the evolutionary change from traditional green computing towards modern cloud-native sustainability.

B. Internal Empirical Study (Anonymized): This was complemented by a literature study and an empirical data gathering phase in a large financial institution. This phase made it possible to identify particular operational hurdles like legacy systems and stringent financial security requirements. These practitioner insights were critical to providing the context for ensuring that proposed guidelines are "operationalizable" within a very regulated environment.

Internal Resources: A review of the institution's internal architectural standards, infrastructure provisioning logs, and ESG reporting frameworks was performed.

Table 1: Systematic Literature Search Parameters

Database	Search Query / Parameters
Springer	("architecture" AND "compute sustainability" AND ("cloud" OR "enterprise" OR "platform"))
IEEE Xplore	("architecture" OR "architectural guideline") AND ("compute" OR "cloud-native" OR "platform engineering") AND ("sustainability" OR "energy efficiency" OR "carbon-aware")
ACM Digital	("architecture guideline" OR "software architecture") AND ("sustainable compute" OR "energy-efficient cloud" OR "carbon-aware computing")

Expert Interviews: Semi-structured interviews were conducted with five key stakeholders to capture diverse perspectives on compute resource management:

- **Solution Architects:** To understand application-level resource demands.
- **Product Managers:** To align sustainability goals with business value and delivery timelines.
- **Enterprise Engineers:** To identify infrastructure-level impediments to right-sizing and resource reuse.

The insights gathered from these practitioners provided the necessary context to ensure the proposed guidelines are "operationalizable" within a highly regulated environment.

4.5 Application of selection criteria

To ensure the relevance and quality of the literature included in this study, a formal set of Inclusion (IC) and Exclusion (EC) criteria was applied. These criteria were designed to filter the initial search results, focusing specifically on the intersection of sustainable compute infrastructure and architectural guidelines.

It follows, then, that designing the criteria for literature selection would result in research that was both technically thorough and chronologically pertinent for this study, running throughout the transformational period from 2016 to 2025. In filtering studies that explicitly investigate PUE, Carbon Intensity, and resource optimization within regulated sectors, thus this approach fills perceived gaps between high-level ESG ambitions and platform engineering operational realities. The ensuing guidelines are, therefore, not of a theoretical nature but are to be "operationalizable" within a high-availability corporate fintech environment where environmental sustainability is taken as a first-class Non-Functional Requirement.

4.6 Snowballing

To complement the systematic database search, a snowballing approach was employed to ensure comprehensive coverage of sustainable architectural guidelines. This process involved two distinct phases:

Table 2: Selection Criteria for Literature (2016-2025)

Category	ID	Criterion Description	Rationale
Inclusion	IC1	Studies from 2016-2025.	Captures modern cloud-native practices.
	IC2	Architectural patterns for efficiency.	Informs the proposed guidelines.
	IC3	PUE or Carbon Intensity metrics.	Answers RQ1 on technical metrics.
	IC4	Financial or regulated sectors.	Ensures viability for banking mandates.
Exclusion	EC1	Published before 2016.	Avoids obsolete infrastructure models.
	EC2	Focus on hardware or e-waste.	Focus is on software and platforms.
	EC3	Non-English or abstracts only.	Ensures full-text data integrity.
	EC4	Generic ESG policy only.	Maintains focus on engineering (RQ2).

- **Forward Snowballing:** Identifying newer papers (up to 2025) that cited the primary selected studies to capture the most recent advancements in carbon-aware scheduling and platform engineering.
- **Backward Snowballing:** Reviewing the reference lists of the initial papers to locate earlier foundational frameworks or metrics, such as the original GHG Protocol or early Green IT standards, that remain relevant to financial ESG compliance.

This hybrid search methodology—systematic database queries combined with iterative snowballing ensures that the resulting guidelines are based on both stable industry standards and cutting edge technological innovations in green platform engineering. Hence, the study establishes, by validation of academic results using empirical data derived from expert interviews in the field of technology, a genuine way to pursue operationalizing sustainability to high-availability financial environments concerning technical feasibility.

4.7 Data Extraction

To answer RQ1 and RQ2, the information necessary to conduct this work was systematically derived. Indeed, in using a standardized Companion Data Extraction Spreadsheet, the likely analysis carried out would be consistent and reproducible for literature from ACM, IEEE, and Springer published between 2016 and 2025

Extraction Process This process of filling in the companion spreadsheet developed through three rigorous steps:

- **Selection and Review :** Full review of each paper that passed the selection criteria was carried out. Abstract, Methodology, and Conclusion are prioritized in this review to bring out Actionable Architectural Guidelines.
- **Categorization of Attributes:** Data was categorized in some columns of the spreadsheet according to the different research questions. Technical data regarding PUE and Carbon Intensity was extracted for RQ1, while qualitative

descriptions on Right-sizing and Multi-tenancy were collected for RQ2.

Conflict Resolution In case a paper contained ambiguous technical information, it was re-examined upon cross-checking its internal citations (backward snowballing) to ensure that the data entered into the spreadsheet was accurate.

Table 3: Data Extraction Attributes for Companion Spreadsheet

Attribute Group	Specific Data Fields	Objective
Bibliographic	Title, Authors, Year (2016–2025), Source Database.	To ensure traceability and confirm temporal relevance.
Technical (RQ1)	PUE, Carbon Intensity Metrics, Energy Consumption Data.	To define the technical metrics required for green SOPs.
Operational (RQ2)	Right-sizing Methods, Multi-tenancy Patterns, Security Overheads.	To identify principles for sustainable platform engineering.
Industrial Context	Financial Sector Constraints, Latency Requirements.	To ensure the guidelines are "operationalizable" in banking.

4.8 Data Synthesis

After completing the extraction of the data, the information for the findings encompassed in the main studies was processed and summarized using the Thematic Synthesis methodology. The main aim was to derive a set of cohesive architectural guidelines directed toward a financial technology department from diverse technical findings.

Synthesis Stages: The synthesis was carried out in three iterative stages:

- **Coding and Categorization:** Each data entry in the accompanying spreadsheet is given a "code" based on its main technical contribution. Thus, data entries dealing with CPU limits are coded under Resource Optimization. , while those regarding regional grid carbon intensity are coded under Carbon-Aware Infrastructure.
- **Theme Development:** Related codes are grouped under higher-order themes directly addressing the research questions. Technical metrics from RQ1. PUE, Carbon Intensity were synthesized to form a "Measurement Framework," while operational concepts RQ2. Right-sizing, Reuse were synthesized to form "Architectural Pillars."
- **Guideline Formulation:** The last themes were made more specific into guidelines for action. A cross-comparison was then made between the academic best practices and constraints discovered during the internal empirical study, such as high availability and security overheads, to test "operationalizability."

Summarization Method: To give a straightforward overview of the extracted data, the synthesis will be summarized in a Mapping Table which relates the architectural principles found with the technical metrics that underpin them and the particular research questions they respectively address.

Table 4: Summary of Data Synthesis: Themes and Guidelines

Synthesized Theme	Architectural Guideline	RQ Link
Technical Codification	Implement real-time monitoring of Carbon Intensity and PUE within CI/CD pipelines.	RQ1
Workload Optimization	Apply strict Right-sizing and auto-scaling to eliminate idle "zombie" resources.	RQ2
Infrastructure Efficiency	Prioritize Multi-tenant cluster reuse over dedicated instance provisioning.	RQ2
Security-Energy Balance	Optimize encryption and auditing logic to minimize computational overhead.	RQ2

4.9 Study Replicability

All supporting resources used for this study are available in my on-drive https://github.com/Victor1207/2025_Olasehinde_Victor-Architecture-Guideline-For-In-company-Technology-Dept-.git, to allow transparency of this study and for others to conduct a similar study and replicate its findings. The replication package comprises:

Research Protocol: The formal plan for the study, including details on the search strings run in ACM, IEEE, and Springer, snowballing, and synthesis steps in detail.

Complete List of Primary Studies: A full bibliographic list of the studies identified from the 2016 to 2025 searches, including those excluded during title/abstract and full text screening.

Classification Framework Parameters: These are definitions and parameters for how to classify the literature provides details of specific technical thresholds for PUE, criteria for identifying "Right-sizing," "Reuse-before-build" patterns.

4.10 Threats to Validity

Harmoniously with the strict criteria characterizing an SLR and the high demands for compliance typical of financial enterprise architecture, this section discusses the threats that could affect the validity of our preliminary investigation. Assuming as a reference the scheme proposed by [1], since the very beginning of the study design, we tried to consider all these aspects in order to have a reliable SoAW.

4.10.1 External validity. This concerns the application possibilities that our results have when generalized to different types of industries.

- Threat: The most significant threat is that our architectural guidelines derived from the synthesis could fit just a bit too well within the realm of hyperscale cloud systems, possibly derailing their universality when it comes to the application domain within existing, on-site financial core systems.
- Mitigation: This has been alleviated by ensuring our included studies encompass a range from edge computing [16] to public cloud systems [17]. The universality is retained through our concentration on high-level architectural principles, which aren't dependent on vendor-dependent code.

4.10.2 Internal validity. Internal validity deals with research design quality and the possibility of systemic bias.

- Threat: The research itself may be biased if the researcher chooses the 'green' strategies inadequately or combines them inadequately, resulting in an incomplete representation of the state of the art.
- mitigation: To overcome the above threat, we strictly followed a pre-specified search procedure on the most popular digital libraries. We were particularly interested in works that employed experimental validation at an industry level [7] to ensure that the selected guidelines were not based on assumptions but on factual knowledge.

4.10.3 Construct validity. It verifies how well the conceptual model (for example, "Sustainability" or "Operational Efficiency") has been captured in the research.

- Threat: The absence of a commonly agreed "Digital Sufficiency" term [9] may result in architectural requirements being interpreted loosely.
- Mitigation: To counter this issue, we matched different literature nomenclature to the Enterprise Architecture Management (EAM) common vocabulary. This helps ensure harmony in the preliminary research constructs employing the same vocabulary in financial governance.

4.10.4 Conclusion validity. Concerns the reliability of the derived relationships between architectural design and energy outcomes.

- Threat: Traditional metrics usually cannot capture the "opaque" interactions in cloud-native stacks ; it may draw wrong conclusions on the efficacy of certain tactics.
- Mitigation: our study explicitly points at the gaps in service-level energy modeling.

By naming the limitations, we set a conservative and realistic baseline for the SoAW, highlighting how automated experimentation frameworks are needed to validate architectural outcomes.

5 RESULTS

This section reports the synthesis of current literature, structured to inform the baseline requirements for the upcoming Architectural Guidelines.

5.1 Mapping the "Technology Depth" Taxonomy

The literature suggests that "technology depth" in modern enterprises is no longer a linear stack but a multi-dimensional ecosystem. Our synthesis identifies three primary clusters of maturity:

- Hardware-Centric Efficiency: Traditional focus on data center Power Usage Effectiveness (PUE).
- Virtualization & Orchestration: The "dark side" of cloud provisioning where resource over-allocation leads to "zombie instances" [16].
- Cognitive & Service Depth: The emerging layer where AI/ML and microservice meshes introduce complex energy profiles that are currently "invisible" to standard monitoring.

5.2 Publication Trends: From Theory to Experimentation

The review reveals a critical pivot point in 2024. While earlier publications were largely conceptual, recent works [7, 18] provide **empirical benchmarks** for tactics like service-level modeling. This trend validates that the field has reached sufficient maturity to support the creation of formal in-company guidelines.

5.3 Comparative Performance of Architectural Patterns

A key result for our guideline development is the performance variance between architectural choices. Research into REST frameworks and execution environments [5] demonstrates that resource consumption can vary by over 40% depending on the stack choice, providing a clear "evidence-based" mandate for internal standardization.

6 DISCUSSION

The discussion evaluates how these findings translate into actionable "Architectural Guidelines" within a high-compliance corporate environment.

6.1 The "Sustainability vs. Performance" Trade-off

The most significant discussion point for the SoAW is the myth that sustainability degrades performance. The literature indicates that many Green Architectural Tactics such as database query optimization and efficient payload serialization actually improve latency and reduce operational costs [17]. For an in-company guideline, this means sustainability can be framed as Performance Engineering.

6.2 Potential for Industrial Adoption (The "Friction" Analysis)

Despite the benefits, the potential for adoption is currently hindered by a lack of "developer-centric" tooling. Funke et al. [7] highlight that if a tactic adds significant complexity to the CI/CD pipeline, it will be ignored. Therefore, our guidelines must focus on automation and transparency rather than manual checklists.

6.3 Strategic Roadmap for the SoAW

As a preliminary study, this investigation confirms that the **In-Company Technology Depth** guidelines must move toward a "Digital Sufficiency" model [9]. This involves:

- Standardizing the Stack: Using the resource profiles found in the results to recommend specific "Golden Paths" for developers.
- Automated Governance: Shifting from annual audits to real-time energy observability at the service level.
- ESG Reporting Integration: Converting technical efficiency gains into metrics understandable to non-technical stakeholders in the financial institution.

7 CONCLUSION

With the increasing emphasis of financial firms to enhance their tech depth integration of high-performance computing and AI-based services, there is an imminent and integral need to make their "technology depth needed to meet sustainable challenges" an operational reality. This research study has emphasized and highlighted the essential integration of architectural principles to make sure that all internal tech development at banks is highly competent and effective and meets all means of upcoming ESG reporting principles. This work employed a systematic mapping approach in which we examined a set of papers from 2016 to 2025 from prominent academic and industry repositories. By examining both theoretical architecture strategies and actual industry assessments, we have determined that our study has captured the "state of the art" in terms of energy efficient design, service-level modeling, and infrastructure assessment for cloud native enterprise infrastructure.

Findings indicate the presence of a large "transparency gap" within the existing architecture of cloud-native systems, where standard infrastructure observability values, such as CPU usage, neglect the complicated energy costs of inter-container service communication and secondary observability systems. Moreover, it was found within the aforementioned research that, within high compliance environments, the application of many environmentally friendly architectural methods is hindered by the absence of standard governance and proven measurement methods. One of the main implications of such a study is the need to have sustainability formalized via the enterprise architecture life cycle within a Statement of Architectural Work (SoAW). By embracing the concept of reusable classification proposed in this paper, financial institutions can reduce risks related to resource consumption. As future work, pilot-testing of these recommendations is scheduled within a production level Kubernetes environment towards maturing the concept of "carbon-aware" standard operating procedures (SOPs) to enable transition from a "digital sufficiency" paradigm, where handling technical and environmental complexity is considered a single conceptual dimension.

8 ACKNOWLEDGMENT

Declaration on Generative AI: I acknowledge the use of digital tools, including ChatGPT v5.2 and WriteFull, for language editing support and not content generation, which assisted in improving the clarity, structure, presentation, and articulation of this paper

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